

Muan Li

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EDUCATION

North Carolina State University, Raleigh, NC

2025 - Present

- Ph.D. in Chemical & Biomolecular Engineering (expected 2030), GPA 3.71/4.00. Advisor: Prof. Alexa Easley. Focus: controlled polymerization of redox-active polymers; electrochemical characterization.

China University of Petroleum-Beijing

2021 - 2025

- B.Eng. in Chemical Engineering, GPA 4.12/5.00, Rank 1/59. National Scholarship (2022, 2023, 2024; top 0.2% nationwide).

RESEARCH EXPERIENCE

Graduate Research Assistant, NC State University

- Developing metered-addition RAFT polymerization of 10-(4-vinylbenzyl)phenoxazine (VBPX), a redox-active styrenic monomer, to achieve precise control over molecular weight distributions via regulated chain-transfer agent (CTA) feeding.
- Formulated fed-batch metered-addition protocols by coupling experimental ^1H NMR kinetics with continuous syringe-pump dosing, decoupling instantaneous monomer concentration from bimolecular termination to tailor custom MWD profiles.
- Characterize redox polymers via cyclic voltammetry (CV), EIS, and chronoamperometry to correlate macromolecular structure with electrochemical performance.

Undergraduate Researcher, China University of Petroleum-Beijing

- Oxygen evolution electrocatalysis: synthesized Ce-doped NiFe-LDH catalysts (170 mV overpotential @ 10 mA cm^{-2}); validated stability in an anion-exchange-membrane water electrolyzer; performed DFT calculations correlating enhanced catalytic activity with optimized adsorption energy of oxygen intermediates.
- CO_2 electroreduction: built Cu-In bimetallic electrodes on CNT-modified carbon paper; secured project funding from the National College Students' Innovation Training Program.
- Heterogeneous electro-Fenton: co-precipitated Ag/ Fe_3O_4 on activated carbon fibers via one-pot hydrothermal synthesis for hydroxyl radical generation. Developed an integrated photoelectrocatalytic reactor for oilfield wastewater treatment.

PROJECTS

RAFT Polymerization Kinetic Simulator (Python)

github.com/muan-li/raft-metered-addition

- Method-of-moments ODE model of RAFT kinetics with metered CTA feeding; predicts MWD evolution under varied feed profiles to guide experimental design.

TECHNICAL SKILLS

Programming & modeling: Python (NumPy, SciPy, Matplotlib), MATLAB; Aspen Plus; Materials Studio (DFT); Blender; Origin.

Characterization: NMR, GPC, FT-IR, CV/EIS/CA, XRD, SEM/TEM, TGA/DSC, GC, UV-Vis.

Languages: Mandarin (native), English (fluent).